VIKRAM EJJAGIRI



Industrial AI · Machine Learning · Digital Production

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PROFILE

Master's student in Applied AI for Digital Production Management (M.Eng.) at TH Deggendorf with a Bachelor's in Mechanical Engineering (Honors in Robotics). I build industrial AI end to end: from machine data collection (OPC UA, MQTT) to predictive maintenance, visual quality inspection and an LLM-based maintenance copilot. **Available immediately** for a working student position, an internship, a Master's thesis or a full-time role.

EDUCATION

03/2025 – present

M.Eng. Applied AI for Digital Production Management
Technische Hochschule Deggendorf, Germany
Current grade: 2.3 (German scale, 1.0 = best) · 90 ECTS, taught in English
Modules: Machine Learning & Deep Learning in Production and Logistics · Advanced Statistical Methods & Optimization (SPC, design of experiments) · Production Data Management, ERP & MES · Big Data, MLOps & NLP · Cyber-Physical Production Systems & Simulation (AnyLogic) · Production and Logistics Management (Lean, Six Sigma) · Quality & Sustainability (FMEA, process capability) · AI and production-systems case studies

08/2021 – 05/2024

B.Eng. Mechanical Engineering (Honors in Robotics)
Vasavi College of Engineering, Hyderabad, India
CGPA 8.17/10 · Bachelor's thesis: Modified SCARA robot · Honors thesis: Design and analysis of an ornithopter

2018 – 2021

Diploma in Mechanical Engineering
Government Polytechnic, Warangal, India
CGPA 9.52/10

PROJECTS

Kaizen Matrix: my own industrial AI project series (2026), each with code, tests and documentation on GitHub. Prototypes on simulated or public data, not yet validated on a real line

Production Digitalization Platform: Sensor to Dashboard — MQTT · TimescaleDB · Docker · scikit-learn · SimPy
github.com/vikram-ejjagiri-ai/production-digitalization-platform

- End-to-end shopfloor platform with one-click start (Docker Compose): real compressor data (MetroPT-3) and a simulated assembly line with RFID (SimPy) over MQTT into TimescaleDB, with harmonization of messy station data and RFID de-duplication.
- KPI engine for OEE, FPY, MTBF/MTTR and RFID lead time; health model (Isolation Forest + SPC z-score) trained on the commissioning period: in the end-to-end run the alarm opened about 2.5 h before the first reported failure.
- AI advisor with retrieval over FMEA/SOPs and a local LLM (rule-based fallback); three Grafana dashboards as code, SQL views for Power BI; 20 unit tests.

Kaizen Assist: GenAI Knowledge Assistant for Engineering — RAG · Text-to-SQL · DuckDB · Ollama · Streamlit
github.com/vikram-ejjagiri-ai/kaizen-assist

- Hybrid retrieval (BM25 + embeddings, RRF) over specifications, test procedures and PDF test reports with version awareness; cited answers from a local LLM (Ollama) or an API.
- Guarded Text-to-SQL on DuckDB (semantic layer, sqlglot guard, read-only) and a compare tool that checks measured values against the released specification limit.
- Guardrails against prompt injection, safety bypass and guessing; evaluation: 96.9 % on 64 development questions, 75 % on a second domain without tuning; 25 unit tests.

OEE Dashboard & Maintenance Copilot Python · pandas · Streamlit · Ollama · RAG

github.com/vikram-ejjagiri-ai/kaizen-oee-maintenance-copilot

- OEE engine following ISO 22400 with data-quality checks (duplicate MES records, wrong product booking); KPIs tested automatically against an independent calculation.
- Analysed 13 weeks of (simulated) MES data: plant OEE 76.8 %, biggest losses from speed loss and changeovers, night-shift die changes on the press taking 73 instead of 40 minutes.
- RAG copilot with a local LLM (Qwen 2.5 7B via Ollama): numbers come from the OEE engine, hybrid search (BM25 + embeddings) over manuals and work orders, safety guardrails; 95.9 % pass rate on 49 test questions.

Visual Quality Inspection with Anomaly Detection PyTorch · ONNX Runtime · Streamlit

github.com/vikram-ejjagiri-ai/kaizen-visual-quality-inspection

- Implemented PatchCore from scratch in PyTorch, trained only on defect-free images (MVTec AD, metal nut).
- Image AUROC 1.000 (baseline 0.828), pixel AUROC 0.989; with a cost-based threshold caught 46 of 47 unseen defects with zero good parts rejected.
- Operator review app with heatmaps and audit log; ONNX export, about 120 ms per image on CPU.

Predictive Maintenance: Failure Classification for a CNC Milling Machine scikit-learn · SHAP · Power BI · DAX

github.com/vikram-ejjagiri-ai/kaizen-machine-failure-prediction

- Public synthetic dataset AI4I 2020: failure classification, not remaining-useful-life prediction (no timestamps).
- Physics-based features (power, temperature difference, tool strain) cut the model's error cost by about a third.
- Random forest with 82 % recall at 93 % precision, benchmarked against expert rules; every alarm explained with SHAP; Power BI dashboard (machine health, alarms, model performance, threshold what-if).

Industry 4.0 Data Platform OPC UA · MQTT · PostgreSQL · Grafana

github.com/vikram-ejjagiri-ai/kaizen-factory-stack

- Simulated bottling line with 7 sensors as an OPC UA server; fault-tolerant OPC UA to MQTT bridge in a Unified Namespace.
- Storage in PostgreSQL and a live Grafana dashboard with vibration zones based on ISO 10816.

SCARA Robot for Multi-Plane Operations (Bachelor's thesis) Fusion 360 · AutoCAD · MSC Adams

- Redesigned the mechanical structure and positioning system for 3D movement; design validation and dynamic simulation of motion and positioning accuracy.

SKILLS

Programming & data	Python, SQL, pandas, NumPy, Git, Pytest
Machine learning & vision	scikit-learn, PyTorch, SHAP, anomaly detection, ONNX
LLMs	RAG, Text-to-SQL, Ollama (local LLMs), hybrid search, guardrails, LLM evaluation, Streamlit
Industry 4.0	OPC UA, MQTT, PostgreSQL/TimescaleDB, Docker Compose, Grafana, Power BI/DAX, OEE, RFID, MES and maintenance data
Design & automation	SolidWorks, Fusion 360, AutoCAD, MSC Adams, PLC (basic), ROS (basic)
Languages	English (professional) · German (B1, telc certificate 2026) · Telugu (native)

FURTHER INFORMATION

Leadership	President of the Robotics Club, Vasavi College of Engineering (from 12/2023); organised line-follower and maze-solver robot workshops
Award	State rank 43 in the Engineering Common Entrance Test (ECET), 2020
Training	Advanced Robotics – NPTEL (2024) · Design of HVAC Systems in Pharmaceutical Industry – KG MECH (2022) · Industrial training on lathe machines – IRA OM Technologies Pvt. Ltd. (2021)